

Radar Velocimeter
LD306S Product
Brochure

Shenzhen Hi-Link
Electronics Co.V1.5

catalogs

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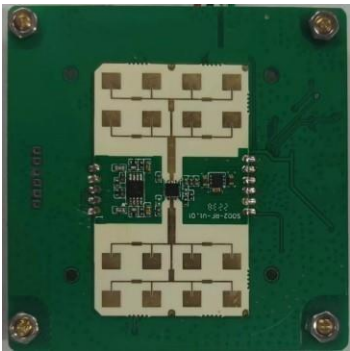
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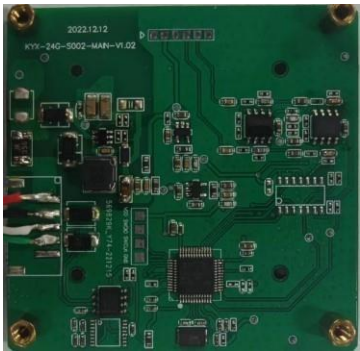
Summarize

LD306S is a radar module for detecting the speed of vehicle movement

1、Appearance and Interface



Module Antenna Surface



modular digital surface

The interface wiring sequence is shown in the table below

NO.	Color	PCBA silkscreen	function
1	Red	+12V	+12V Input
2	Blue	GND	earth (wire)
3	Green	TX/A	RS232 Transmit /RS485A+
4	Yellow	RX/B	RS232 Receiving /RS485B-

2、Technical Parameters

Parameters	Clarification
frequency of emission	24.125GHz
firing power	40dBm
Antenna Style	Horizontal Angle 45° /Vertical Angle 90°
Operating Current	70mA
operating voltage	DC12V
operating temperature	-40~+80℃
data interface	RS232/485 serial port
data format	9600, 8, n, 1
response time	≥60ms
Detection range	10~250km/h
Detection distance	≤150m
Detection accuracy	1km/h
Product Size	64mm*65mm

3、Data Protocols

Communication configuration: R485 serial port, default baud rate 9600, data bit 8, stop bit 1, no check bit. The module defaults to automatic reporting mode, and continuously reports the traveling speed of the vehicle under power-on condition.

Communication format:

Uplink data (sent by module)

character threshold	header	lengths	address	speed	Debugging data	checking
byte count	2	1	1	2	6	1
content	55A5	0A	D3	XX	XX...XX	XX

Explanation:

Report data information, the reporting interval can be set from 60ms to 2s, the default is 400ms, and each word threshold is analyzed as follows:

- Length: number of bytes except frame header and parity byte, 0x0A, fixed byte
- Address: 0xD3, fixed byte speed: unit: km/h, occupies 2 bytes, high bit first
- Debug data: debugging information, accounting for 6 bytes
- Checksum: sum of all bytes except the threshold of the checksum word, and take the lower 8 bits, accounting for 1 byte.

eg:

55 A5 0A D3 00 58 00 01 01 4C 01 01 7F

The speed of the car is 88 km/h.

- Downlink data (received by module)

character threshold	header	address	command number	command parameter	Checking	end of frame
byte length	2	1	1	2	1	2
content	BAAB	00	XX	XX	00	55BB

Description:

- Frame header: fixed 2 bytes, 0xBAAB
- Address: 0x00, fixed byte
- Command No.: Setting parameter command number, accounts for 1 byte.
- Command Parameter: Set parameter value, occupies 2 bytes.
- Parity: no parity, 0x00, fixed byte
- End of frame: fixed 2 bytes, 0x55BB
- Setting successfully returned: 0D 0A 77 72 69 74 65 20 6F 6B 0D 0A

That's ASCII code "enter + write ok + enter".

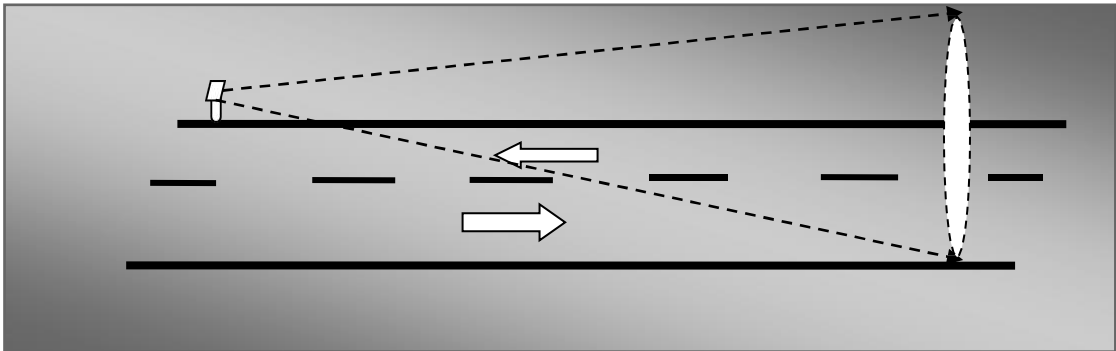
The parameter setting command table is as follows:

**Common
Instruction
List 1**

Function	Command number (1 byte)	Command Parameters (2 bytes)	description
Minimum detection speed	E0	Value range: 10~200, unit: km/h, default is 10km/h.	Sets the minimum speed detected by the module
Max detection speed	E5	Value range: 10~350, unit: km/h, default is 250 km/h.	Set the maximum speed detected by the module
(level of) sensitivity	E1	Value range: 1~20, unit: gear, default: 4th gear	Sets the detection sensitivity of the module; the smaller the value of the parameter, the farther the detection distance.

communications protocol	F6	Value 6, 7 6: Protocol 6, query mode 7: Protocol 7, auto-reporting mode		The communication data format and mode can be modified by modifying the protocol type.
baud	D4	retrieve value baud		Setting the communication baud rate
		48	4800	
		96	9600	
		384	38400	
		576	57600	
		1152	115200	
		default (setting):	9600	
output target	D9			Select the target for the output speed
		value	Output	
		0	latest car	
		1	fastest car	
		Default: 1, most recent vehicle speed		

4、Installation of applications



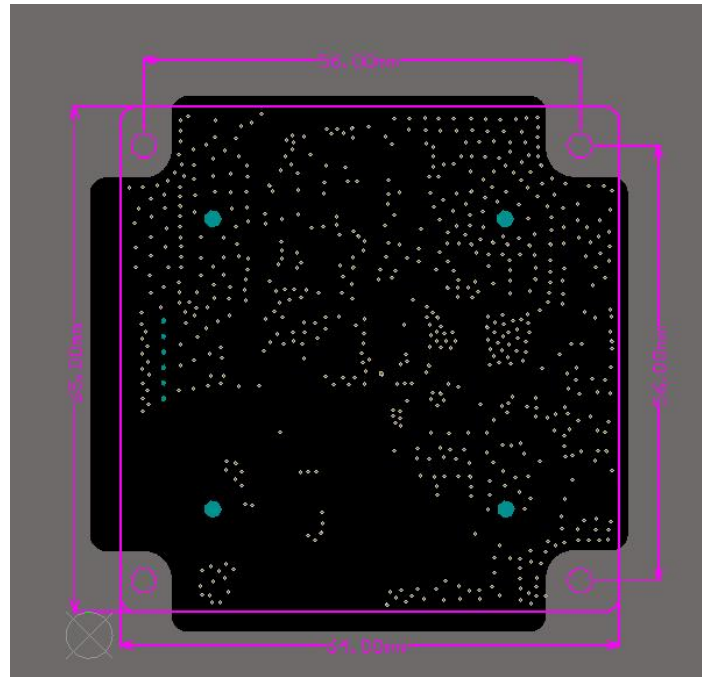
Installation Diagram of Speed Measuring Radar

- Installed at the roadside, height 2m, radar vertical road surface, detection range as the dotted line in the schematic diagram area
- When mounted in a billboard or at a height of 6m such as a flyover, tilt the radar 3° toward the road surface.



5、Mechanical dimensions

PCBA dimension drawing mm



6、Precaution

- (1) This module only detects relatively moving targets within the detection area, completely stationary targets are not detected. The output is the speed and other information of the nearest/fastest target in relative motion within the detection area.
- (2) The module defaults to continuous auto-reporting (protocol 7). For query mode, set the protocol type to 6 and send a hexadecimal query command (55 5A 02 D3 84).
- (3) If the module designs its own shell, the distance between the radar antenna surface and the shell surface is the best 6mm, the thickness of the shell is about 2~3mm, the antenna surface shell material can not use metal or gold-plated metal paint material, it is recommended to use PC or plastic, steel, etc., and the back of the antenna is made of metal material.